

Problems with constraints of ' $\geq$ ' and ' $=$ '

Until now it was assumed that the problem was in the in the standard form, that is:

$$\max z = \mathbf{c}'\mathbf{x}$$

subject to

$$\mathbf{Ax} \leq \mathbf{b}$$

$$\mathbf{x} \geq \mathbf{0}$$

$$\text{and } \mathbf{b} \geq \mathbf{0}$$

For other legitimate forms of LP model (with constraints $\geq$ or $=$), some adjustments have to be made so that problems can be solved by the Simplex method.

The only real problem with these new forms, is to identify an **initial feasible basic solution**.

Until now, this solution was determined, quite conveniently, as follows: initially the original variables were the **NBV** (zero value) and the "slack" variables were the **BV** (the value of the respective constraint independent term).

Now, the approach to use is the **artificial variable** technique.

This technique constructs a new **auxiliary problem** by introducing a false variable (called artificial variable) for each constraint in which it is required. This new variable is added only with the purpose of being the initial **BV** for that constraint.

The non-negativity constraints are also needed for the artificial variables. The objective function is altered by them.

Constraints of ‘=’

Any equality constraint:

$$a_{i1}X_1 + a_{i2}X_2 + \dots + a_{in}X_n = b_i$$

is equivalent to the pair of constraints:

$$a_{i1}X_1 + a_{i2}X_2 + \dots + a_{in}X_n \leq b_i$$

$$a_{i1}X_1 + a_{i2}X_2 + \dots + a_{in}X_n \geq b_i$$

However, instead of doing this replacement, which increases the number of constraints, we use the **artificial variable** technique.

Consider the following example:

$$\text{Maximize } z = 5 x_1 + 2 x_2$$

Subject to

$$x_1 \leq 3$$

$$x_2 \leq 4$$

$$x_1 + 2 x_2 = 9$$

$$x_1 \geq 0, x_2 \geq 0$$

After the introduction of the “slack” variables (necessary for the constraints of type $\leq$), the problem in the enlarged form is obtained:

$$\text{Maximize } z = 5 x_1 + 2 x_2$$

Subject to

$$x_1 + x_3 = 3$$

$$x_2 + x_4 = 4$$

$$x_1 + 2 x_2 = 9$$

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0$$

It is found that these equations do not have an obvious initial feasible basic solution, because there is not a "slack" that can be used as **BV** of the 3rd equation.

The technique of the artificial variable overcomes this difficulty by introducing a nonnegative artificial variable (x_5) in this equation, just like a "slack".

Thus, the 3rd equation is transformed in:

$$x_1 + 2 x_2 + x_5 = 9$$

with the non-negativity constraint, $x_5 \geq 0$.

Proceeding in the usual way, we get a basic feasible solution to the problem:

$$\mathbf{x} = (0, 0, 3, 4, 9)$$

Constraints of '≥'

The direction of an inequality is always reversed when both sides are multiplied by -1.

As a result, any functional constraint of type "≥" may be converted to an equivalent constraint of type '≤', multiplying it by -1.

Consider the following example:

$$\text{Maximize } z = 5 x_1 + 2 x_2$$

Subject to

$$x_1 \leq 3$$

$$x_2 \leq 4$$

$$x_1 + 2 x_2 \geq 9$$

$$x_1 \geq 0, x_2 \geq 0$$

The 3rd constraint can be converted into one of type ≤, in the following way:

$$x_1 + 2 x_2 \geq 9 \quad \Leftrightarrow$$

$$\Leftrightarrow -x_1 - 2 x_2 \leq -9$$

The problem in the enlarged form, with the “slack” variables x_3, x_4 e x_5 , is thus:

$$\text{Maximize } z = 5 x_1 + 2 x_2$$

Subject to

$$x_1 + x_3 = 3$$

$$x_2 + x_4 = 4$$

$$-x_1 - 2 x_2 + x_5 = -9$$

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0, x_5 \geq 0$$

However, remind that the Simplex method assumes that all independent terms are non-negative ($b \geq 0$).

It is this assumption that allows you to select the variables "slack" to be the initials BV (with value equal to the independent term of the respective constraint) and, thus, to obtain an initial feasible basic solution.

However, a negative independent term, such as the one of the 3rd constraint, yields a negative value for the "slack" variable x_5 , in the initial solution: $x = (0, 0, 3, 4, -9)$. That is, this solution does not respect the non-negativity constraint of x_5 .

Multiplying the 3rd constraint by -1 the independent term becomes a positive value, but the coefficient of x_5 changes to -1: $x_1 + 2x_2 - x_5 = 9$. That is, the value of this variable remains negative in the initial solution.

However, the constraint in this format can be viewed as an equality constraint with a non-negative independent term. Thus, the technique of artificial variable can be applied, such as was done previously.

If x_6 is the non-negative artificial variable for this constraint, the final form is

$$x_1 + 2x_2 - x_5 + x_6 = 9$$

where variable x_6 is used as initial **BV** and x_5 as **NBV**.

Therefore, the initial **FBS** for this problem would be:

$$\mathbf{x} = (0, 0, 3, 4, 0, 9)$$

The variable x_5 is named “surplus” (excess), because it subtracts the excess of the left side in relation to the right side of the constraint, in order to convert the inequality into an equivalent equation.

Conclusion

The artificial variables are added to the constraints of types '=' and ' $\geq$ ' in order to serve as initial BV for these same constraints.

However, these variables have no physical meaning, they are just used as a mathematical stratagem. For this reason, they should not be included in the optimal solution of the problems with a non-zero value. In order to guarantee this condition, two techniques can be used:

- “**Big M**” technique
- “**Two Phases**” technique

Concluding:

Type of constraint	Type of variable to add	Characteristics
$\leq$	slack	➤ Coefficient 1 in the constraint in which it is added ➤ Excess of the value of b in relation to the left side of the constraint ➤ Coefficient 0 in the objective function
$\geq$	surplus	➤ Coefficient -1 in the constraint in which it is added ➤ Excess of the left side of the constraint in relation to b ➤ Coefficient 0 in the objective function
$\geq e =$	artificial	➤ Coefficient 1 in the constraint in which it is added ➤ It doesn't have a physical meaning so it can not appear in the optimal solution: – “Big M” technique – “Two Phases” technique ➤ The coefficient in the objective function depends on the used technique (“Big M” or “two Phases”)

”Big M” technique

In this technique, artificial variables are strongly penalized in the objective function so as to cause their fast nulling.

This penalty consists in assigning them an arbitrarily large and negative coefficient ($-M$).

Since the Simplex method works in order to improve the objective function value, it will naturally tend to remove the artificial variables from the basis.

”Two Phases” technique

The problem is solved in two phases:

1st Phase - Consists of removing the artificial variables from the basis in order to obtain a feasible basic solution with only real variables of the problem: this will be the 1st **FBS** for the initial problem and will be the starting solution for the second phase.

2nd Phase – Starts from the **FBS** obtained in the previous phase and applies the Simplex method to achieve the optimal solution.

Example 1

Consider that in Example 1 presented in Chapter II, was imposed the full cultivation of the land area of 4, and the employment of at least 9 people.

The mathematical model is now:

Determine

x_1 = area to plant of rice

x_2 = area to plant of corn

in order to

maximize $z = 5 x_1 + 2 x_2$

subject to

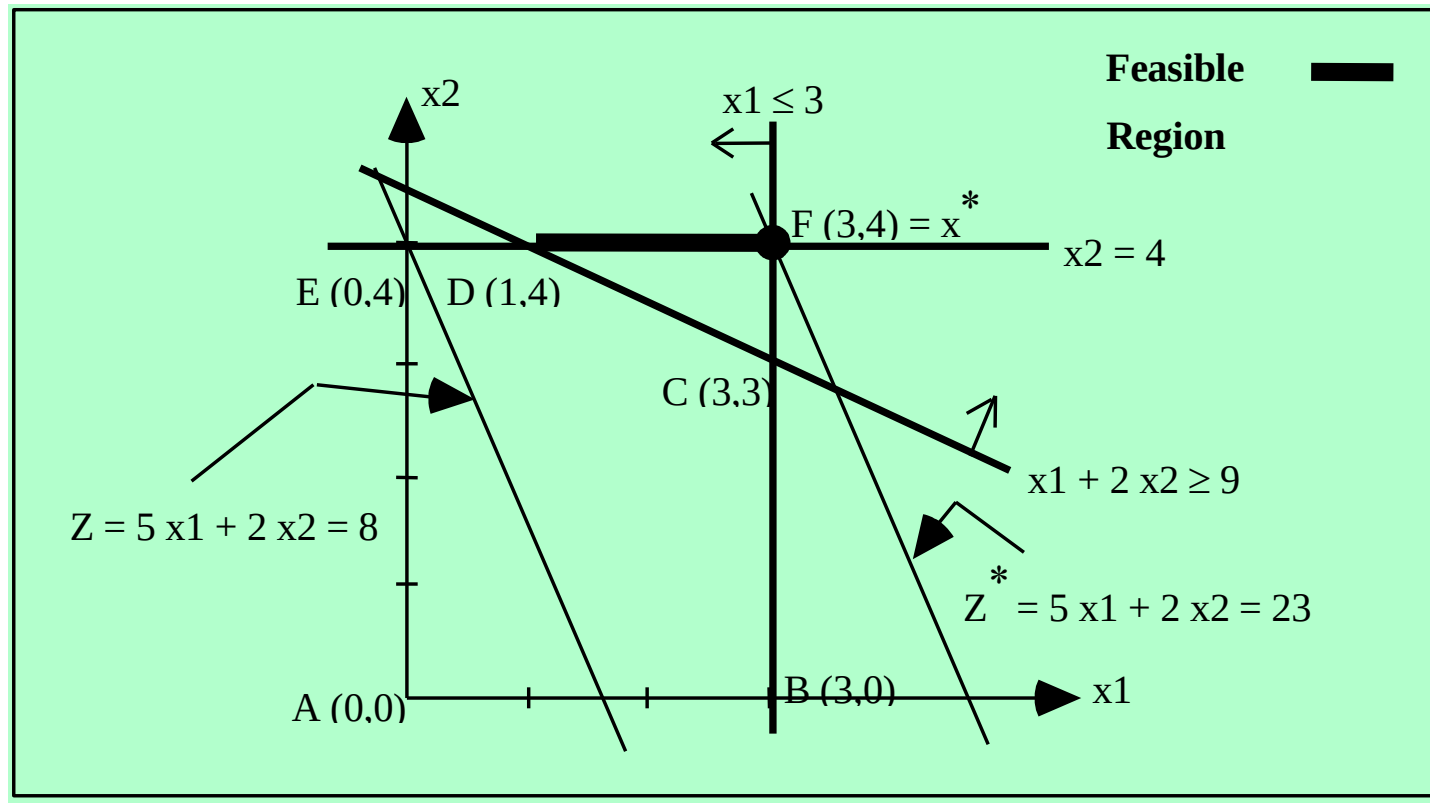
$x_1 \leq 3$ (area to plant of rice)

$x_2 = 4$ (area to plant of corn)

$x_1 + 2 x_2 \geq 9$ (hand labour)

$x_1 \geq 0, x_2 \geq 0$

Solving by the graphical method



Solving by the Simplex method with the “Big M” technique

Determine

$$\mathbf{x} = (x_1, x_2, x_3, x_4, x_5, x_6) \geq \mathbf{0}$$

in order to

$$\text{maximize } z = 5x_1 + 2x_2 - Mx_5 - Mx_6$$

$\downarrow \quad \downarrow$
artificial

subject to

$$x_1 + x_4 = 3 \quad (1)$$

$\uparrow$
slack

$$x_2 + x_5 = 4 \quad (2)$$

surplus *artificial*
 $\downarrow \quad \downarrow$

$$x_1 + 2x_2 - x_3 + x_6 = 9 \quad (3)$$

$$x_i \geq 0, i = 1, 2, \dots, 6$$

Initial table

C_i		5	2	0	0	-M	-M	
x_B	C'_B	x_1	x_2	x_3	x_4	x_5	x_6	b
x_4	0	1	0	0	1	0	0	3
$\leftarrow x_5$	-M	0	<u>1</u> *	0	0	1	0	4
x_6	-M	1	2	-1	0	0	1	9
$Z_j - C_j$		-5	-2	M	0	0	0	-13M
		-M	-3M					
			$\uparrow$					

Feasible basic solution for the auxiliary problem:

BVs	NBVs	
$x_4 = 3$	$x_1 = 0$	
$x_5 = 4$	$x_2 = 0$	
$x_6 = 9$	$x_3 = 0$	$z = -13M$

Corresponds to POINT **A** $\rightarrow$ not feasible for the initial problem

C_i		5	2	0	0	-M	-M	
x_B	C'_B	x_1	x_2	x_3	x_4	x_5	x_6	b
x_4	0	1	0	0	1	0	0	3
x_2	2	0	1	0	0	1	0	4
$\leftarrow x_6$	-M	<u>1</u> *	0	-1	0	-2	1	1
$Z_j - C_j$		-5	0		0	2	0	8
		-M		M		3M		-M
		$\uparrow$						

Feasible basic solution for the auxiliary problem:

BVs

$$x_4 = 3$$

$$x_2 = 4$$

$$x_6 = 1$$

NBVs

$$x_1 = 0$$

$$x_3 = 0$$

$$x_5 = 0$$

$$z = 8 - M$$

Corresponds to POINT **E** $\rightarrow$ not feasible for the initial problem

C_i		5	2	0	0	-M	-M	
x_B	C'_B	x_1	x_2	x_3	x_4	x_5	x_6	b
$\leftarrow x_4$	0	0	0	<u>1</u> *	1	2	-1	2
x_2	2	0	1	0	0	1	0	4
x_1	5	1	0	-1	0	-2	1	1
$z_j - C_j$		0	0	-5	0	8	5	13
						M	M	

↑

Feasible basic solution for the initial problem (x_5 and x_6 are **NBV**s):

BVs	NBVs	
$x_4 = 2$	$x_3 = 0$	
$x_2 = 4$	$x_5 = 0$	
$x_1 = 1$	$x_6 = 0$	$z = 13$

Corresponds to POINT **D** → feasible for the initial problem but not optimal

C_i		5	2	0	0	-M	-M	b
x_B	C'_B	x_1	x_2	x_3	x_4	x_5	x_6	
x_3	0	0	0	1	1	2	-1	2
x_2	2	0	1	0	0	1	0	4
x_1	5	1	0	0	1	0	0	3
$z_j - C_j$		0	0	0	5	2	0	23
						M	M	

Optimal table because there are no negative values in the $z_j - c_j$ row.

Optimal solution:

BVs

$$x_1^* = 3$$

$$x_2^* = 4$$

$$x_3^* = 2$$

NBVs

$$x_4^* = 0$$

$$x_5^* = 0$$

$$x_6^* = 0$$

The optimal solution is therefore to plant all the land and to employ 11 people, with a maximum profit of 23 UM.

with $z^* = 23$

Corresponds to POINT **F** → feasible and optimal

Solving by the Simplex method with the “Two Phases” technique

1st Phase

In this step the coefficients of the objective function are all zero except for the artificial variables that are -1.

Determine

$$\mathbf{x} = (x_1, x_2, x_3, x_4, x_5, x_6) \geq \mathbf{0}$$

in order to

$$\text{maximize } z_{1\text{st Phase}} = \underset{\substack{\uparrow \\ \text{artificial}}}{-x_5} - \underset{\substack{\uparrow \\ \text{artificial}}}{x_6}$$

subject to

$$x_1 + \underset{\substack{\uparrow \\ \text{slack}}}{x_4} = 3 \quad (1)$$

$$x_2 + \underset{\substack{\uparrow \\ \text{artificial}}}{x_5} = 4 \quad (2)$$

$$\underset{\substack{\downarrow \\ \text{surplus}}}{x_1} + 2x_2 - x_3 + \underset{\substack{\downarrow \\ \text{artificial}}}{x_6} = 9 \quad (3)$$

$$x_i \geq 0, i = 1, 2, \dots, 6$$

Initial table (of 1st phase)

C_i		0	0	0	0	-1	-1	
C'_B		x_1	x_2	x_3	x_4	x_5	x_6	b
x_4	0	1	0	0	1	0	0	3
← x_5	-1	0	<u>1</u> *	0	0	1	0	4
x_6	-1	1	2	-1	0	0	1	9
$Z_j - C_j$		-1	-3	1	0	0	0	-13

↑

Feasible basic solution for the auxiliary problem:

BVs

$$x_4 = 3$$

$$x_5 = 4$$

$$x_6 = 9$$

NBVs

$$x_1 = 0$$

$$x_2 = 0$$

$$x_3 = 0$$

$$Z_{1^{afase}} = -13$$

Corresponds to PONTO **A** → not feasible for the initial problem

C_i		0	0	0	0	-1	-1	
x_B	C'_B	x_1	x_2	x_3	x_4	x_5	x_6	b
x_4	0	1	0	0	1	0	0	3
x_2	0	0	1	0	0	1	0	4
$\leftarrow x_6$	-1	<u>1</u> *	0	-1	0	-2	1	1
$Z_j - C_j$		-1	0	1	0	3	0	-1

↑

Feasible basic solution for the auxiliary problem:

BVs

$$x_4 = 3$$

$$x_2 = 4$$

$$x_6 = 1$$

NBVs

$$x_1 = 0$$

$$x_3 = 0$$

$$x_5 = 0$$

$$Z_{1st \text{ Phase}} = -1$$

Corresponds to POINT **E** $\rightarrow$ not feasible for the initial problem

C_i		0	0	0	0	-1	-1	b
x_B	C'_B	x_1	x_2	x_3	x_4	x_5	x_6	
x_4	0	0	0	1	1	2	-1	2
x_2	0	0	1	0	0	1	0	4
x_1	0	1	0	-1	0	-2	1	1
$Z_j - C_j$		0	0	0	0	1	1	0

Optimal table of first phase because there are no negative values in row $z_j - c_j$.

Optimal solution of the 1st phase (first feasible basic solution for the initial problem since x_5 and x_6 are **NBV**s):

BVs	NBVs	
$x_4 = 2$	$x_3 = 0$	
$x_2 = 4$	$x_5 = 0$	
$x_1 = 1$	$x_6 = 0$	$Z_{1^{st} \text{ phase}} = 0$

Corresponds to PONTO **D** → feasible for the initial problem

2nd Phase

In this step the coefficients of the objective function are the coefficients of the vector $\mathbf{c}'$ of the proposed problem, namely

$$\max z_{2\text{nd phase}} = \max z = 5 x_1 + 2 x_2$$

Initial table (of 2nd phase)

$\mathbf{c}_i$		5	2	0	0	
$\mathbf{x}_B \mathbf{c}'_B \mathbf{x}_i$		$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_4$	$\mathbf{b}$
$\leftarrow \mathbf{x}_4$	0	0	0	<u>1</u> *	1	2
$\mathbf{x}_2$	2	0	1	0	0	4
$\mathbf{x}_1$	5	1	0	-1	0	1
$z_j - \mathbf{c}_j$		0	0	-5	0	13
		$\uparrow$				

The **FBS** corresponds to POINT **D** $\rightarrow$ feasible but not optimal

$$Z_{2\text{nd phase}} = 13$$

ci		5	2	0	0	
$\mathbf{x_B} \mathbf{c}' \mathbf{B}^{-1} \mathbf{x_i}$		$\mathbf{x_1}$	$\mathbf{x_2}$	$\mathbf{x_3}$	$\mathbf{x_4}$	$\mathbf{b}$
$\mathbf{x_3}$	0	0	0	1	1	2
$\mathbf{x_2}$	2	0	1	0	0	4
$\mathbf{x_1}$	5	1	0	0	1	3
$Z_j - c_j$		0	0	0	5	23

The **BFS** corresponds to POINT **F** → feasible and optimal

Optimal solution:

BVs

$$x_1^* = 3$$

$$x_2^* = 4$$

$$x_3^* = 2$$

NBVs

$$x_4^* = 0$$

$$x_5^* = 0$$

$$x_6^* = 0 \quad z_{2nd\ phase}^* = z^* = 23$$

As it was concluded by the “Big M” technique, the optimal solution is to plant all the land and to employ 11 people, with a maximum profit of 23 UM.

Example 2

A small company manufactures three types of electronic kits (A, B and C).

The net profit of each unit is of 5 monetary units (MU) for kits A, 10 MU for Kits B, and 15 MU for kits C.

The time available limits the total number of kits that can be produced to 500.

Market studies show that the total number of kits types A and B should be at least 100.

The number of kits of type A should exceed exactly in 120 units the total number of kits types B and C.

How many units of kits of types A, B and C shall be produced in order to maximize profit?

Therefore, the problem consists in:

Determine

x_1 = number of kits of type A to be produced

x_2 = number of kits of type B to be produced

x_3 = number of kits of type C to be produced

in order to

$$\text{maximize } z = 5 x_1 + 10 x_2 + 15 x_3$$

subject to

$$x_1 + x_2 + x_3 \leq 500$$

$$x_1 + x_2 \geq 100$$

$$x_1 - x_2 - x_3 = 120$$

$$\mathbf{x} = (x_1, x_2, x_3) \geq \mathbf{0}$$

Solving by the Simplex method with the “Big M” technique

Determine

$$\mathbf{x} = (x_1, x_2, x_3, x_4, x_5, x_6, x_7) \geq \mathbf{0}$$

in order to

$$\max z = 5x_1 + 10x_2 + 15x_3 - M \underset{\substack{\uparrow \\ \text{artificial}}}{x_6} - M \underset{\substack{\uparrow \\ \text{artificial}}}{x_7}$$

subject to

$$x_1 + x_2 + x_3 + x_5 = 500 \quad (1)$$

$$x_1 + x_2 - \underset{\substack{\uparrow \\ \text{surplus}}}{x_4} + \underset{\substack{\uparrow \\ \text{artificial}}}{x_6} = 100 \quad (2)$$

$\downarrow$

$$x_1 - x_2 - x_3 + x_7 = 120 \quad (3)$$

$$x_i \geq 0, i = 1, 2, \dots, 7$$

Inicial table

c_i	5	10	15	0	0	-M	-M	
$x_B \quad c'_B \quad x_i$	x_1	x_2	x_3	x_4	x_5	x_6	x_7	b
$x_5 \quad 0$	1	1	1	0	1	0	0	500
$\leftarrow x_6 \quad -M$	<u>1</u> *	1	0	-1	0	1	0	100
$x_7 \quad -M$	1	-1	-1	0	0	0	1	120
$z_j - c_j$	-5	-10	-15		0	0	0	
	-2M		M	M				-220M
	↑							

BVs

$$x_5 = 500$$

$$x_6 = 100$$

$$x_7 = 120$$

NBVs

$$x_1 = 0 \quad x_4 = 0$$

$$x_2 = 0$$

$$x_3 = 0$$

$$z = -220M$$

C_i	5	10	15	0	0	-M	-M	
$x_B \ C'_B \ x_i$	x_1	x_2	x_3	x_4	x_5	x_6	x_7	b
$x_5 \ 0$	0	0	1	1	1	-1	0	400
$x_1 \ 5$	1	1	0	-1	0	1	0	100
$\leftarrow x_7 \ -M$	0	-2	-1	<u>1</u> *	0	-1	1	20
$Z_j - C_j$	0	-5	-15	-5	0	5	0	500
		2M	M	-M		2M		-20M
				↑				

C_i	5	10	15	0	0	-M	-M	
$x_B \ C'_B \ x_i$	x_1	x_2	x_3	x_4	x_5	x_6	x_7	b
$\leftarrow x_5 \ 0$	0	2	<u>2</u> *	0	1	0	-1	380
$x_1 \ 5$	1	-1	-1	0	0	0	1	120
$x_4 \ 0$	0	-2	-1	1	0	-1	1	20
$Z_j - C_j$	0	-15	-20	0	0		5	600
						M	M	
			↑					

		c_i	5	10	15	0	0	-M	-M		
		$x_B \quad c'_B \quad x_i$	x_1	x_2	x_3	x_4	x_5	x_6	x_7	b	
x_3	15		0	1	1	0	0.5	0	-0.5	190	
x_1	5		1	0	0	0	0.5	0	0.5	310	
x_4	0		0	-1	0	1	0.5	-1	0.5	210	
$z_j - c_j$			0	5	0	0	10		-5	4400	
								M	M		

Optimal table because there are no negative values in row $z_j - c_j$.

Optimal solution:

BVs

$$x_3 = 190$$

$$x_1 = 310$$

$$x_4 = 210$$

NBVs

$$x_2 = 0$$

$$x_5 = 0$$

$$x_6 = 0$$

$$x_7 = 0$$

$$z^* = 4400$$

(From the 3rd table, x_6 e x_7 are **NVBs** → the obtained basic solutions are feasible for the initial problem)

Solving by the Simplex method with the “Two Phases” technique

1st Phase

In this step the coefficients of the objective function are all zero except for the artificial variables that are **-1**.

Determine

$$\mathbf{x} = (x_1, x_2, x_3, x_4, x_5, x_6, x_7) \geq \mathbf{0}$$

in order to

$$\max z_{1^{st} \text{ phase}} = - \underset{\substack{\uparrow \\ \text{artificial}}}{x_6} - \underset{\substack{\uparrow \\ \text{artificial}}}{x_7}$$

subject to

$$x_1 + x_2 + x_3 + \underset{\substack{\uparrow \\ \text{slack}}}{x_5} = 500 \quad (1)$$

$$x_1 + x_2 - \underset{\substack{\uparrow \\ \text{surplus}}}{x_4} + \underset{\substack{\uparrow \\ \text{artificial}}}{x_6} = 100 \quad (2)$$

$$x_1 - x_2 - x_3 + \underset{\substack{\downarrow \\ \text{artificial}}}{x_7} = 120 \quad (3)$$

$$x_i \geq 0, i = 1, 2, \dots, 7$$

Initial table (of 1st phase)

C_i	0	0	0	0	0	-1	-1	
$x_B \ C'_B \ x_i$	x_1	x_2	x_3	x_4	x_5	x_6	x_7	b
$x_5 \ 0$	1	1	1	0	1	0	0	500
$\leftarrow x_6 \ -1$	<u>1</u> *	1	0	-1	0	1	0	100
$x_7 \ -1$	1	-1	-1	0	0	0	1	120
$Z_j - C_j$	-2	0	1	1	0	0	0	-220
	↑							

C_i	0	0	0	0	0	-1	-1	
$x_B \ C'_B \ x_i$	x_1	x_2	x_3	x_4	x_5	x_6	x_7	b
$x_5 \ 0$	0	0	1	1	1	-1	0	400
$x_1 \ 0$	1	1	0	-1	0	1	0	100
$\leftarrow x_7 \ -1$	0	-2	-1	<u>1</u> *	0	-1	1	20
$Z_j - C_j$	0	2	1	-1	0	2	0	-20
				↑				

ci		0	0	0	0	0	-1	-1	b
$x_B \ C'_B \ x_i$		x_1	x_2	x_3	x_4	x_5	x_6	x_7	
x_5	0	0	2	2	0	1	0	-1	380
x_1	0	1	-1	-1	0	0	0	1	120
x_4	0	0	-2	-1	1	0	-1	-1	20
$Z_j - c_j$		0	0	0	0	0	1	1	0

Optimal table of 1st phase

Optimal solution of the 1st phase:

BVs

$$x_5 = 380$$

$$x_1 = 120$$

$$x_4 = 20$$

NBVs

$$x_2 = 0$$

$$x_3 = 0$$

$$x_6 = 0$$

$$x_7 = 0$$

Variables x_6 e x_7 are **NBVs** → feasible solution for the initial problem

2nd Phase

In this step the coefficients of the objective function are the coefficients of the vector $\mathbf{c}'$ of the proposed problem, namely

$$\max z_{2\text{nd phase}} = \max z = 5x_1 + 10x_2 + 15x_3$$

Initial table (of 2nd phase)

		$\mathbf{c}_i$	5	10	15	0	0	$\mathbf{b}$
$\mathbf{x}_B$	$\mathbf{c}'_B$	$\mathbf{x}_i$	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_4$	$\mathbf{x}_5$	
$\leftarrow \mathbf{x}_5$	0		0	2	<u>2</u>	0	1	380
$\mathbf{x}_1$	5		1	-1	-1	0	0	120
$\mathbf{x}_4$	0		0	-2	-1	1	0	20
$Z_j - \mathbf{c}_j$			0	-15	-20	0	0	600
					↑			

$\mathbf{C_i}$	5	10	15	0	0	
$\mathbf{x_B \ C' B^x_i}$	$\mathbf{x_1}$	$\mathbf{x_2}$	$\mathbf{x_3}$	$\mathbf{x_4}$	$\mathbf{x_5}$	$\mathbf{b}$
$\mathbf{x_3 \ 15}$	0	1	1	0	0.5	190
$\mathbf{x_1 \ 5}$	1	0	0	0	0.5	310
$\mathbf{x_4 \ 0}$	0	-1	0	1	0.5	210
$\mathbf{Z_j - C_j}$	0	5	0	0	10	4400

Optimal table of the 2nd phase

Optimal solution:

BV_s

$$x_3 = 190$$

$$x_1 = 310$$

$$x_4 = 210$$

NBV_s

$$x_2 = 0$$

$$x_5 = 0$$

$$z^* = 4400$$